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First order necessary conditions for a local minimum

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- The necessary condition of **Theorem 1** is not sufficient for a local minimum

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- Check that f has a **global minimum** at $x_1 = \frac{1}{2}, x_2 = 0$

Second order necessary conditions for a point to be a local minimum

- **Theorem 3:** Let $f : \Omega \to \mathbb{R}$ be a **twice continuously differentiable** function (that is all the second order partial derivatives of f (given by $\frac{\partial^2 f}{\partial x_j \partial x_i}$) exists and are continuous as functions from Ω to $\mathbb{R}$).

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- **Definition:** A real symmetric matrix A is said to be positive semidefinite (negative semidefinite) if $\mathbf{x}^T A\mathbf{x} \geq 0$ ($\mathbf{x}^T A\mathbf{x} \leq 0$) for all $\mathbf{x} \in \mathbb{R}^n$.

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- For example the $\mathbf{0}$ matrix has all n eigenvalues equal to 0, the identity matrix I_n has all n eigenvalues equal to 1, and for an **upper triangular matrix**, the diagonal entries are its eigenvalues.

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- **Remark:** A nonsingular (nonzero determinant) **positive semidefinite matrix** is **positive definite**.

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  Hence f is **not** a **convex function** on Ω.

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- **Remark :** Since minimizing f is same as maximizing $-f$, all the previous theorems for **minimizing** a **convex function** have corresponding analogues for **maximizing** a **concave function.**

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- **Theorem 6:** Let f be a **convex function** defined on a **closed** and **bounded convex set** Ω (so it has atleast one extreme point), then there exists an extreme point of Ω, where f takes its **maximum** value.